

HW1

4

We can simply create a z^* as a strict convex converge of x^*, y^* .

Here to create 5 optimal z^* , let $\lambda = [\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{5}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{7}}]$ and $z^* = \lambda x^*, (1 - \lambda)y^*$ creates 5 non integer solutions. Using irrational λ guarantees a non-integer solution

5

a

Let variables $x_1, x_2, \dots, x_m$ be binary LP variables to represent the n -arcs in our set A . A variable a_i is equal to 1 iff it is chosen as a part of our subset and it does not conflict with other variables in our subset and 0 otherwise. Since we are trying to find the largest subset, our objective function should be to maximize $\sum_i x_i$. To ensure no conflicting, we create n constraints, with each constraint corresponding to a number $1 - n$ (problem definition was unclear if we include 0 in our arcs but if that is the case simply add a new 0 constraint). For these constraints, x_i has a coefficient of 1 iff it's arc covers the integer corresponding to the constraint (which can easily be found by iterating through the integers each arc covers).

Thus our binary LP looks like this:

$$\max \sum_i x_i$$

$$\text{s.t. } Ax = b$$

$$\forall i, x_i \in \{0, 1\}$$

$b = 1_n$ which is a vector of length n of all 1's

And to formulate A :

$$a_{ij} = \begin{cases} 0 & \text{if the } j\text{th arc does not cover } i \\ 1 & \text{if the } j\text{th arc covers } i \end{cases}$$

b

$$A = \{(1, 1), (1, 2), (1, 3), (2, 4), (2, 5), (2, 6), (2, 7), (2, 8), (3, 9)\}$$

c

d

$$A = \{(1, 2), (2, 3), (3, 4), (4, 6), (5, 7), (6, 8), (7, 9), (8, 1), (9, 3)\}$$

e

6

a

Suppose there exists an optimal solution S with arc a_r appearing in two districts d_1, d_2 . Create an S' by removing a_r from d_2 , this is still a valid solution because all arcs are being represented and removing an arc from a district cannot introduce any sort of conflicting arcs. We also know that S and S' are both optimal solutions because the number of districts did not change from S to S' and we defined S as the optimal solution. We can repeat this process recursively until we end with an S^* with each arc appearing a single time and similarly to the S' idea, S^* is also an optimal solution because no new districts were created.

b

Omit as solved in HW2 **5.a**

c

$$A = \{(1, 1), (1, 2), (1, 3), (2, 4), (2, 5), (2, 6), (2, 7), (2, 8), (3, 9)\}$$

d

e

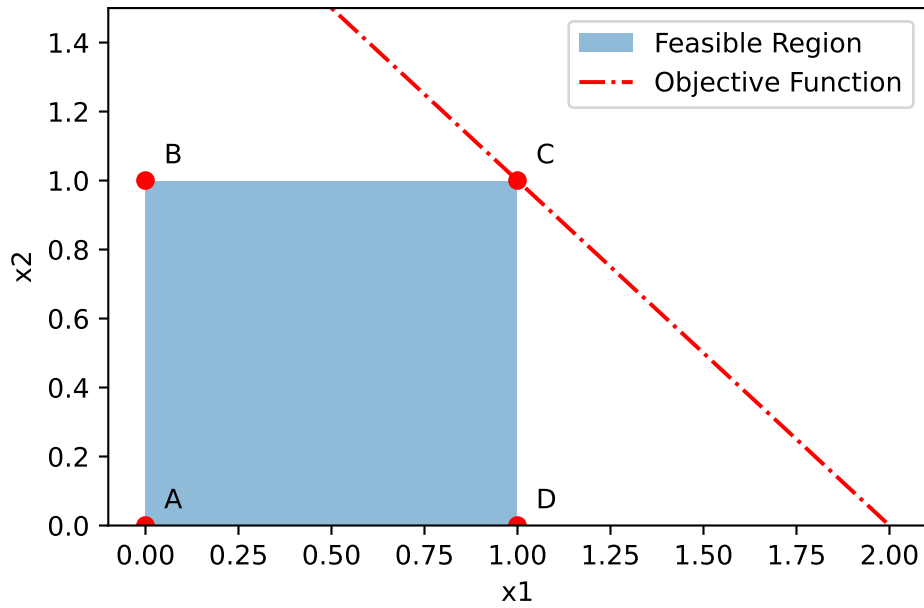
$$A = \{(1, 2), (2, 3), (3, 4), (4, 6), (5, 7), (6, 8), (7, 9), (8, 1), (9, 3)\}$$

f

HW2

1

a



b

Remember the standard form is (from HW1 with slight changes for consistency):

$$\min -x_1 - x_2$$

$$\text{s.t. } 2(x_1) + (x_2) + s_3 = 3$$

$$x_1 + s_1 = 1$$

$$x_2 + s_2 = 1$$

$$x_1, s_1, x_2, s_2, s_3 \geq 0$$

So we can write A as:

$$A = \begin{pmatrix} 2 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix}, C = \begin{pmatrix} -1 \\ -1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Thus we can create our B and L for the origin as:

$$B = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad L = \begin{pmatrix} 2 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Thus we have:

$$\pi = (B^{-1})^T C_B = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}^T \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

And it trivially follows that:

$$C^\pi = C - A^T \pi = \begin{pmatrix} -1 \\ -1 \\ 0 \\ 0 \\ 0 \end{pmatrix} - A^T \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

c

$$\text{A BFS for the optimal point is } \bar{x} = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Thus we can create our B and L for the optimal point $\bar{x}$ as:

$$B = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad L = \begin{pmatrix} 0 & 1 \\ 0 & 0 \\ 1 & 0 \end{pmatrix}$$

Thus we have:

$$\pi = (B^{-1})^T C_B = \begin{pmatrix} 0.5 & 0 & -0.5 \\ 0 & 0 & 1 \\ -0.5 & 1 & 0.5 \end{pmatrix}^T \begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.5 & 0 & -0.5 \\ 0 & 0 & 1 \\ -0.5 & 1 & 0.5 \end{pmatrix} \begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -0.5 \\ 0 \\ -0.5 \end{pmatrix}$$

And it trivially follows that:

$$C^\pi = C - A^T \pi = \begin{pmatrix} -1 \\ -1 \\ 0 \\ 0 \\ 0 \end{pmatrix} - A^T \begin{pmatrix} -0.5 \\ 0 \\ -0.5 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ 0 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} -1 \\ -1 \\ 0 \\ -0.5 \\ -0.5 \end{pmatrix} = \begin{pmatrix} -2 \\ -2 \\ 0 \\ -0.5 \\ -0.5 \end{pmatrix}$$

d

We can find the basic solutions through the following formula then check if the solution is a feasible one:

$$b = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}, x_B = \begin{pmatrix} B^{-1}b \\ 0 \end{pmatrix}$$

Naively there are 5 choose 3 different basis but there are a few basis that are invalid as they dont span R^3 . Namely these were the basis that span R^3

$$B_1 = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, B_2 = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}, B_3 = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, B_4 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$$B_5 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, B_6 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, B_7 = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, B_8 = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Thus we can find all of the B^{-1} :

$$B_1^{-1} = \begin{pmatrix} 0.5 & 0 & -0.5 \\ 0 & 0 & 1 \\ -0.5 & 1 & 0.5 \end{pmatrix}, B_2^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -2 & 0 \\ -1 & 2 & 1 \end{pmatrix}, B_3^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -2 & -1 \end{pmatrix}, B_4^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

$$B_5^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, B_6^{-1} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{pmatrix}, B_7^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -2 & 0 \end{pmatrix}, B_8^{-1} = \begin{pmatrix} 0.5 & 0 & 0 \\ -0.5 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Let x^i denote the basic solution for basis B_i

$$x^1 = \begin{pmatrix} 1 \\ 1 \\ -1 \\ 0 \\ 0 \end{pmatrix}, x^2 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, x^3 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, x^4 = \begin{pmatrix} 0 \\ 3 \\ 1 \\ -2 \\ 0 \end{pmatrix}, x^5 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 3 \end{pmatrix}, x^6 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \\ 2 \end{pmatrix}, x^7 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}, x^8 = \begin{pmatrix} 1.5 \\ 0 \\ -0.5 \\ 1 \\ 0 \end{pmatrix}$$

Of these it is clear that x^1, x^4, x^8 are not feasible solutions.

e

Omit, moved to HW3

f

Omit, moved to HW3

2

a

b

i

ii

iii

3

a

b

c

Omit, moved to HW3

d

Omit, moved to HW3

e

4

a

$$\min s_1 + s_2$$

$$\text{s.t. } 2x_1 + x_2 + s_3 = 3$$

$$x_1 + s_1 = 1$$

$$x_2 + s_2 = 1$$

$$x_1, x_2, s_1, s_2, s_3 \geq 0$$

b

We can convert this into standard form by adding vector s such that $|x| = |s|$ and we add new constraints: $x + s = u$ and of course $x, s \geq 0$. To create our new cost vector $\tilde{c}$ where $|\tilde{c}| = |x| + |s|$ and we create it using the following piecewise function by flipping the sign of any negative costs and assigning that cost to the corresponding slack variable:

$$\tilde{c}_i = \begin{cases} 0 & \text{if } i \leq |x| \wedge c_i < 0 \\ c_i & \text{if } i \leq |x| \wedge c_i \geq 0 \\ 0 & \text{if } i > |x| \wedge c_{i-|x|} \geq 0 \\ -c_{i-|x|} & \text{if } i > |x| \wedge c_{i-|x|} < 0 \end{cases}$$

Thus our cost vector has only positive components and our new LP is as follows and is an equivalent minimization problem:

$$\begin{aligned} \min \quad & \tilde{c}^T \begin{pmatrix} x \\ s \end{pmatrix} \\ \text{s.t.} \quad & Ax = b \\ & x + s = u \\ & x, s \geq 0 \end{aligned}$$

5

a

Omit, moved to HW3

b

c

d